Problem 2: Merging Two Ticket Counters

TASK

Two railway counters each produce a singly linked list of ticket numbers in ascending order. At closing time, both lists must be combined into one ascending list.

Implement `Node mergeSorted(Node a, Node b)` that merges the lists by relinking the existing nodes and returns the new head. Duplicates are kept. Use your own node class (for example `class Node { int data; Node next; }`); do not use `java.util.LinkedList` or other collections for the list itself.

CONSTRAINTS

- 0 <= length of each list <= 10^5
- No new nodes may be created (a single dummy node is allowed)

EXAMPLES

Sample 1

```
a = 3 -> 8 -> 15
b = 5 -> 8 -> 12 -> 20

Expected Output: 3 -> 5 -> 8 -> 8 -> 12 -> 15 -> 20
```

Sample 2

```
a = (empty), b = 4

Expected Output: 4
```

COMPLEXITY REQUIREMENTS

State the time and space complexity in terms of m and n (the two lengths), and compare it with copying both lists into an array and sorting.

HINT

> *Keep a tail pointer for the merged list and always attach the smaller of the two current nodes; when one list runs out, attach the rest of the other.*
